Step-by-Step Algorithm

University Library Visitor Registration and Daily Counting System

Main program

1. Start the program.
2. Define the constants REGISTER_OPTION = 1, VIEW_OPTION = 2 and EXIT_OPTION = 3.
3. Declare the variables: choice, DailyCount, TotalRecords and i as Integer; StudentName, StudentID, VisitDate and SelectedDate as String; and visitorRecords as an empty list of visit records.
4. Set TotalRecords to 0 and choice to 0.
5. While choice is not equal to 3, repeat steps 6 to 11.
6. Call DisplayMenu to show the three options.
7. Input the operator's choice.
8. If choice = 1, call RegisterVisitor.
9. Otherwise, if choice = 2, call ViewDailyRecords.
10. Otherwise, if choice = 3, display "System Closed".
11. Otherwise, display "Invalid Option". Return to step 5.
12. When choice = 3 the loop ends. End the program.

Module DisplayMenu

1. Display "1. Register Visitor".
2. Display "2. View Daily Records".
3. Display "3. Exit".
4. Return to the main program.

Module RegisterVisitor

1. Input StudentName.
2. Input StudentID.
3. Input VisitDate.
4. Call the function IsValidVisitor with StudentName and StudentID.
5. If the function returns True, save StudentName, StudentID and VisitDate in visitorRecords, add 1 to TotalRecords and display "Visitor Registered Successfully".
6. Otherwise, display "Name and Student ID are required" and do not save the record.

7. Return to the main program.

Function IsValidVisitor(name, id)

1. If name is not blank AND id is not blank, return True.
2. Otherwise, return False.

Module ViewDailyRecords

1. Input SelectedDate.
2. Set DailyCount to 0 and set i to 1.
3. While i is less than or equal to TotalRecords, repeat steps 4 to 6.
4. If the visit date of record i equals SelectedDate, display the record (name, ID and date) and add 1 to DailyCount.
5. Add 1 to i.
6. Return to step 3.
7. If DailyCount = 0, display "No Records Available".
8. Display "Total Visits: " followed by DailyCount.
9. Return to the main program.